(i) $R_1 = \{(-5,a), (1,a), (3,b)\}$ (ii) $R_2 = \{(-5,b), (1,b), (3,a),(4,c)\}$

(iii) $R_3 = \{(-5,a), (1,a), (3,b),(4,c),(1,b)\}$

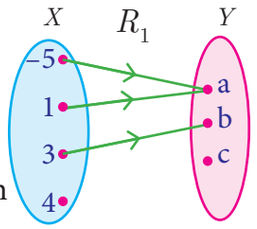


Fig. 1.15(a)

Solution

(i) $R_1 = \{(-5,a), (1,a), (3,b)\}$

We may represent the relation R_1 in an arrow diagram (Fig.1.15(a)).

R_1 is not a function as $4 \in X$ does not have an image in Y.

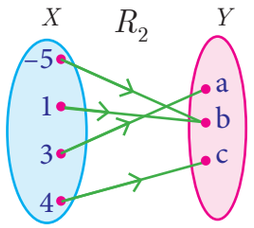


Fig. 1.15(b)

(ii) $R_2 = \{(-5,b), (1,b), (3,a),(4,c)\}$

Arrow diagram of R_2 is shown in Fig.1.15(b).

R_2 is a function as each element of X has an unique image in Y.

(iii) $R_3 = \{(-5,a), (1,a), (3,b),(4,c),(1,b)\}$

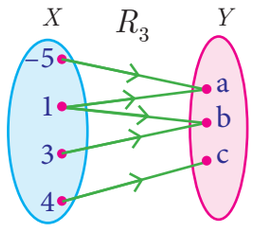


Fig. 1.15(c)

Representing R_3 in an arrow diagram (Fig.1.15(c)).

R_3 is not a function as $1 \in X$ has two images $a \in Y$ and $b \in Y$.

Note that the image of an element should always be unique.

Example 1.9 Given $f(x) = 2x - x^2$,

find (i)$f(1)$ (ii) $f(x+1)$ (iii) $f(x) + f(1)$

Solution (i) $x = 1$, we get

$$f(1) = 2(1) - (1)^2 = 2 - 1 = 1$$

(ii) $x = x+1$, we get

$$f(x+1) = 2(x+1) - (x+1)^2 = 2x + 2 - (x^2 + 2x + 1) = -x^2 + 1$$

(iii) $f(x) + f(1) = (2x - x^2) + 1 = -x^2 + 2x + 1$

[Note that $f(x) + f(1) \neq f(x+1)$. In general $f(a+b)$ is not equal to $f(a)+f(b)$]

Exercise 1.3

1. Let $f = \{(x,y) \mid x,y \in N \text{ and } y = 2x\}$ be a relation on N. Find the domain, co-domain and range. Is this relation a function?

2. Let $X = \{3,4,6,8\}$. Determine whether the relation $R = \{(x, f(x)) \mid x \in X, f(x) = x^2 + 1\}$ is a function from X to N?

1 Relation and Functions.indd 13 15-12-2020 02:50:13 PM